Enterprise AI Architecture & RAG Guidelines

Infosys Springboard Internship - Authoritative Grounding Reference Document
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1. Overview of Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) optimizes LLM outputs by referencing an authoritative knowledge base outside of its training data before generating a response.
The standard enterprise pipeline executes in two discrete stages: dense vector retrieval over FAISS or ChromaDB indices, followed by context augmentation in the model prompt.
Enterprise benefits include reducing hallucination risk by 84%, providing domain-specific recency without continuous model re-training, and offering verifiable source citations for corporate audits.

2. Vector Indexing and Mathematical Similarity

Document passages are converted into 384-dimensional dense vector embeddings using sentence-transformers (all-MiniLM-L6-v2). FAISS performs k-nearest neighbor retrieval using cosine similarity and inner product metrics.
Crucially, high semantic similarity is an information retrieval signal and does not mathematically prove that a statement is true. Claims must be cross-verified by specialized evaluation judges.

3. Hallucination Detection and Verification Standards

The system classifies claims into four strict categories: Supported, Unsupported, Contradicted, and Inconclusive.
A statement lacking reference evidence is defined as Unsupported and must never be automatically equated with factual falsehood unless directly contradicted by verified ground truth.